

# Yu Duan

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## Education

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| Sep 2023 –          | <b>Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology</b> <ul style="list-style-type: none"><li>• Ph.D. Program in Electrical Engineering and Computer Science</li></ul> |
| Sep 2019 – Jun 2023 | <b>Institute for Interdisciplinary Information Sciences, Tsinghua University</b> <ul style="list-style-type: none"><li>• B.Eng. in Computer Science and Technology</li></ul>                                           |
| Feb 2022 – Jul 2022 | <b>Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology</b> <ul style="list-style-type: none"><li>• Visiting Student</li></ul>                                                            |

## Publication

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| 2025 | <b>POCO: Scalable Neural Forecasting through Population Conditioning</b> <ul style="list-style-type: none"><li>• <b>Yu Duan</b>, Hamza Tahir Chaudhry, Misha B. Ahrens, Christopher D Harvey, Matthew G Perich, Karl Deisseroth*, Kanaka Rajan*</li><li>• NeurIPS 2025</li></ul>                                                                                                                                 |
| 2023 | <b>Hebbian and Gradient-based Plasticity Enables Robust Memory and Rapid Learning in RNNs</b> <ul style="list-style-type: none"><li>• <b>Yu Duan</b>, Zhongfan Jia, Qian Li, Yi Zhong, Kaisheng Ma</li><li>• ICLR 2023</li></ul>                                                                                                                                                                                 |
| 2023 | <b>Natural constraints explain working memory capacity limitations in sensory-cognitive models</b> <ul style="list-style-type: none"><li>• Yudi Xie*, <b>Yu Duan</b>*, Aohua Cheng, Pengcen Jiang, Christopher Cueva, Guangyu Robert Yang (*equal contribution)</li><li>• Computational and Systems Neuroscience (COSYNE 2023)</li><li>• Conference on Cognitive Computational Neuroscience (CCN 2022)</li></ul> |

## Research Experience

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| Feb 2024 - Ongoing     | <b>Foundation Models for Neural Dynamics</b> <ul style="list-style-type: none"><li>• Advised by Prof. Kanaka Rajan at Kempner Institute of Harvard University</li><li>• In collaboration with Prof. Karl Deisseroth at Stanford University</li></ul>                                                                                                                               |
| Feb 2022 - Dec 2023    | <b>Human-like Capacity Limitation in Multi-system Models of Working Memory</b> <ul style="list-style-type: none"><li>• Advised by Prof. Robert Yang at the MIT Brain and Cognitive Sciences</li></ul>                                                                                                                                                                              |
| March 2023 - June 2023 | <b>Predicting fMRI Response of Human Visual System with Pre-trained Visual-Textual Neural Networks</b> <ul style="list-style-type: none"><li>• Advised by Prof. Pei Sun at Department of Psychology, Tsinghua University</li></ul>                                                                                                                                                 |
| June 2022 - Sep 2022   | <b>Hebbian and Gradient-based Plasticity Enables Robust Memory and Rapid Learning in RNNs</b> <ul style="list-style-type: none"><li>• Advised by Prof. Kaisheng Ma at Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University</li><li>• Co-advised by Prof. Yi Zhong at IDG/McGovern Institute and School of Life Sciences, Tsinghua University</li></ul> |
| Mar 2021 - Jan 2022    | <b>Modeling the Fly Olfactory System with Plastic and Compartmentalized RNNs</b> <ul style="list-style-type: none"><li>• Advised by Prof. Kaisheng Ma and Prof. Yi Zhong</li></ul>                                                                                                                                                                                                 |

## Honors

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| 2022 | <b>Scholarship for Scientific Innovation, Tsinghua University</b> |
| 2021 | <b>Scholarship for Academic Excellence, Tsinghua University</b>   |
| 2019 | <b>Freshman Scholarship, Tsinghua University</b>                  |
| 2017 | <b>Gold Medal, National Olympiad in Informatics, China</b>        |

## Service

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- **Reviewing:** International Conference on Learning Representations (ICLR)